

Chapter 2 — Finite Markov Decision Processes

Question 1 — Discounted Return

An agent receives the following sequence of rewards:

$$R_{t+1} = 4, \quad R_{t+2} = 3, \quad R_{t+3} = 6, \quad R_{t+4} = 2$$

The discount rate is

$$\gamma = 0.8.$$

Calculate the discounted return G_t .

Response 1

$$G_t = R_{t+1} + \gamma \cdot R_{t+2} + \gamma^2 \cdot R_{t+3} + \gamma^3 \cdot R_{t+4}$$

$$G_t = \sum_{k=0}^3 \gamma^k R_{t+k+1}$$

1. $R_{t+1} = 4$
2. $\gamma \cdot R_{t+2} = 0.8 \times 3 = 2.4$
3. $\gamma^2 \cdot R_{t+3} = (0.8)^2 \times 6 = 0.64 \times 6 = 3.84$
4. $\gamma^3 \cdot R_{t+4} = (0.8)^3 \times 2 = 0.512 \times 2 = 1.024$

Result:

$$G_t = 4 + 2.4 + 3.84 + 1.024 = \mathbf{11.264}$$

Question 2 — Effect of the Discount Factor

An agent receives a reward of 10 after 1 step, 8 after 2 steps, and 6 after 3 steps.

Calculate the return for each of the following discount rates:

1. $\gamma = 0$
2. $\gamma = 0.5$
3. $\gamma = 0.9$

Use:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3}.$$

Which discount rate makes the agent more sensitive to future rewards?

Response 2

1. Case 1: $\gamma = 0$:

$$G_t = 10 + 0 \times 8 + 0^2 \times 6 = 10$$

2. Case 2 $\gamma = 0.5$:

$$G_t = 10 + 0.5 \times 8 + (0.5)^2 \times 6 = 10 + 4 + 0.25 \times 6 = 10 + 4 + 1.5 = 15.5$$

3. Case 3 $\gamma = 0.9$:

$$G_t = 10 + 0.9 \times 8 + (0.9)^2 \times 6 = 10 + 7.2 + 0.81 \times 6 = 10 + 7.2 + 4.86 = 22.06$$

Which discount rate makes the agent more sensitive to future rewards? $\gamma = 0.9$ makes the agent more sensitive to future rewards. As γ approaches 1, future rewards retain a much higher present value.

Question 3 — State-Value Function

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Consider an MDP with two states, A and B . Under policy π :

- From state A , the agent receives a reward of 4 and moves to B with probability 1.
- From state B , the agent receives a reward of 2 and remains in B with probability 1.
- The discount factor is $\gamma = 0.5$.

Using the Bellman equation,

$$v^\pi(s) = E_\pi[R_{t+1} + \gamma v^\pi(S_{t+1}) | S_t = s],$$

calculate:

$$v^\pi(A)$$

and

$$v^\pi(B).$$

Response 3

Calculate $v_\pi(B)$

Since the agent receives a reward of 2 and returns to B ,

$$v_\pi(B) - 0.5v_\pi(B) = 2$$

$$0.5v_\pi(B) = 2$$

$$\boxed{v_\pi(B) = 4}$$

Calculate $v_\pi(A)$

The agent receives reward 4 and then moves to B :

$$v_\pi(A) = 4 + 0.5v_\pi(B)$$

Substitute $v_\pi(B) = 4$:

$$v_\pi(A) = 4 + 0.5(4)$$

$$\boxed{v_\pi(A) = 6}$$

Question 4 — Bellman Equation with Probabilistic Transitions

An agent is in state S_t and follows a fixed policy. The environment behaves as follows:

Transition to S_1 , probability = 0.4, reward = 2

Transition to S_2 , probability = 0.6, reward = 5

$$v_{\pi}(S_1) = 10 \qquad v_{\pi}(S_2) = 20 \qquad \gamma = 0.9$$

Response 4

Next state	Probability	Reward	State value
S_1	0.4	2	10
S_2	0.6	5	20

Discount factor: $\gamma = 0.9$.

The Bellman expectation equation is

$$v_{\pi}(S_t) = \sum_{s', r} p(s', r \mid S_t, a) [r + \gamma v_{\pi}(s')]$$

Step 1: Calculate the contribution from S_1

$$0.4[2 + 0.9(10)]$$

$$= 0.4(11) = 4.4$$

Step 2: Calculate the contribution from S_2

$$0.6[5 + 0.9(20)]$$

$$= 0.6(23) = 13.8$$

Step 3: Add the contributions

$$v_{\pi}(S_t) = 4.4 + 13.8$$

$$\boxed{v_{\pi}(S_t) = 18.2}$$

Question 5 — Action-Value Function

Agent is currently in state S and takes action a

With prob 0.7, reward is 5, and moves to state S_1

With prob 0.3, reward is -2, and moves to state S_2

$\gamma = 0.8$, $(v_\pi(S_1) = 8)$, $(v_\pi(S_2) = 4)$

Response 5

Action-Value Formula:

$$q_\pi(S, a) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = S, A_t = a]$$

$$(0.7): 0.7 \times [5 + 0.8 \times 8] = 0.7 \times 11.4 = 7.98$$

$$(0.3): 0.3 \times [-2 + 0.8 \times 4] = 0.3 \times 1.2 = 0.36$$

$$\text{Total: } q_\pi(S, a) = 7.98 + 0.36 = 8.34$$

Question 6 — Bellman Optimality Equation

Actions available in state S : a_1 and a_2 .

For action a_1 : reward = 4, next state S_1 with prob 1

For action a_2 : reward = 7, next state S_2 with prob 1

$v^*(S_1) = 10, v^*(S_2) = 5, \gamma = 0.9$

Using the Bellman Optimality Equation:

$$v^*(S) = \max_a q^*(S, a)$$

Response 6

Step 1: Evaluate action a_1

$$q^*(S, a_1) = 4 + 0.9 \times 10 = 13$$

Step 2: Evaluate action a_2

$$q^*(S, a_2) = 7 + 0.9 \times 5 = 11.5$$

Step 3: Select the maximum

$$v^*(S) = \max(13, 11.5) = 13$$

Optimal action is a_1

Question 7 — Bellman Optimality with Probabilities

Consider two actions available in state S

$$v^*(S_1) = 12, v^*(S_2) = 8, \gamma = 0.8$$

For action a_1 : $P(S_1|S, a_1) = 0.6$ reward is 4, $P(S_2|S, a_1) = 0.4$ reward is 2

For action a_2 : $P(S_1|S, a_2) = 0.3$ reward is 6, $P(S_2|S, a_2) = 0.7$ reward is 1

Response 7

Step 1: Evaluate action a_1

The transition probabilities are 0.6 to S_1 with reward 4, and 0.4 to S_2 with reward 2.

$$q^*(S, a_1) = 0.6[4 + 0.8(12)] + 0.4[2 + 0.8(8)]$$

Calculate the first branch:

$$0.6(4 + 9.6) = 0.6(13.6) = 8.16$$

Calculate the second branch:

$$0.4(2 + 6.4) = 0.4(8.4) = 3.36$$

Therefore,

$$q^*(S, a_1) = 11.52$$

Step 2: Evaluate action a_2

The transition probabilities are 0.3 to S_1 with reward 6, and 0.7 to S_2 with reward 1.

$$q^*(S, a_2) = 0.3[6 + 0.8(12)] + 0.7[1 + 0.8(8)]$$

Calculate the first branch:

$$0.3(15.6) = 4.68$$

Calculate the second branch:

$$0.7(7.4) = 5.18$$

Therefore,

Step 3: Choose the optimal action

$$v^*(S) = \max(11.52, 9.86)$$

$$v^*(S) = 11.52$$

The optimal action is a_1 .

Question 8 — Recycling Robot

A recycling robot has two states:

$S = \{\text{High}, \text{Low}\}$. $\alpha = 0.8$, $\beta = 0.6$, $r_{\text{search}} = 2$, $r_{\text{wait}} = 1$, $\gamma = 0.9$

Response 8

In state Low, assume that choosing **wait** keeps the robot in the Low state and provides a reward of 1.

Using the Bellman equation:

$$v^*(\text{Low}) = 1 + 0.9v^*(\text{Low})$$

$$0.1v^*(\text{Low}) = 1$$

Therefore,

$$v^*(\text{Low}) = 10$$

In state High, assume that **search** is the optimal action. The robot remains in High with probability $\alpha = 0.8$ and transitions to Low with probability 0.2.

Using the Bellman equation:

$$v^*(\text{High}) = 2 + 0.9[0.8v^*(\text{High}) + 0.2v^*(\text{Low})]$$

Substituting $v^*(\text{Low}) = 10$:

$$v^*(\text{High}) = 2 + 0.72v^*(\text{High}) + 1.8$$

$$0.28v^*(\text{High}) = 3.8$$

Therefore,

$$v^*(\text{High}) \approx 13.57$$

Question 9 — Comparing Two Policies

An MDP has two policies, π_1 and π_2 . Their state-value functions are:

Response 9

State	$v_{\pi_1}(s)$	$v_{\pi_2}(s)$	Better policy
S_1	12	15	π_2
S_2	8	8	Equal
S_3	20	18	π_1

The policy ordering rule is

$$\pi_1 \geq \pi_2 \iff v_{\pi_1}(s) \geq v_{\pi_2}(s) \quad \forall s$$

For $\pi_1 \geq \pi_2$, the condition fails at S_1 , because $12 < 15$.

For $\pi_2 \geq \pi_1$, the condition fails at S_3 , because $18 < 20$.

Therefore,

$$\pi_1 \not\geq \pi_2$$

$$\pi_2 \not\geq \pi_1$$

Question 10 — Optimal Action-Value Function

An MDP has the following action values for state S

$$q^*(S, a_1) = 12.5, q^*(S, a_2) = 15.2, q^*(S, a_3) = 14.7$$

Response 10

$$v^*(S) = \max_a q^*(S, a)$$

$$v^*(S) = \max(12.5, 15.2, 14.7)$$

$$v^*(S) = 15.2$$

The optimal action is

$$a^* = a_2$$

Question 11 — Markov Property

Consider an environment where the next state depends on the current position and current velocity, but not on how the agent reached that position

Response 11

$$S_t = (x_t, v_t) = (5, 2)$$

Two different histories lead to exactly the same state:

History 1: $x = 1 \rightarrow 3 \rightarrow 5$

History 2: $x = 4 \rightarrow 3 \rightarrow 5$

$P(S_{t+1} = S' \mid S_t = (5, 2), A_t = a) = 0.7$ for both histories.

A state representation satisfies the Markov property if the probability distribution of the next state depends only on the current state S_t and action A_t , and not on the past history.

Since the transition probability (0.7) is identical regardless of the history leading to state (5, 2), the state representation (position and velocity) fully summarizes all relevant past information.

Yes, the state representation satisfies the Markov property.